

Session Objectives

- Connect pathophysiology to clinical presentation, laboratory findings, treatment, and management for various anemias of childhood, including iron deficiency anemia, transient erythroblastopenia of childhood, Sickle Cell Anemia and its many complications
- Identify unique and shared characteristics in coagulation issues. Understand clinical presentation, laboratory findings and treatment of Hemorrhagic Disease of the Newborn, Hemophilia and Immune Thrombocytopenia.
- Explain the presentation, symptoms, and laboratory findings of pediatric Leukemias and general approach to treatment.
- Compare and contrast etiology, clinical presentation and laboratory +/- radiographic findings of Neuroblastoma and Wilms tumor, as well as Ewing Sarcoma and Osteosarcoma.
- Outline clinical presentation, diagnosis, and laboratory findings of Retinoblastoma
- Identify complications of pediatric oncology treatment, such as Myelosuppression and Tumor Lysis syndrome.

Pediatric Hematology

- **Anemias:**

- Iron Deficiency Anemia
- Thalassemia
- Aplastic Anemia
- Fanconi Anemia
- Sickle Cell Anemia
- Diamond-Blackfan Anemia
- Physiologic Nadir
- Transient Erythroblastopenia of Childhood (TEC)
- Lead Poisoning
- Hereditary Spherocytosis
- G6PD Deficiency

- **Platelets:**

- Immune Thrombocytopenic Purpura (ITP)
- Hemolytic Uremic Syndrome
- Wiskott-Aldrich Syndrome

- **Bleeding Conditions:**

- Hemophilia
- Von Willebrand Disease
- Hemorrhagic Disease of Newborn

Suspecting Anemia in Pediatrics

- Normal Hemoglobin (Hb) Values vary with age
- Clinical Features: (appear when Hb <8-9g/dL)
 - Chronic:
 - Usually Asymptomatic
 - Mild/ moderate
 - Pallor, weakness, fatigue, irritability
 - Tachycardia, flow murmur
 - Severe or acute
 - SOB on exertion, weakness, tachypnea, high output heart failure, syncope
 - **Red flags** (that not dealing with just Iron Def Anemia)
 - Bone pain, hemolysis, jaundice, splenomegaly, lymphadenopathy
 - Persistent systemic complaints (fever, fatigue, weight loss)
- Initial lab workup:
 - CBC w/ differential, reticulocyte count, Ferritin



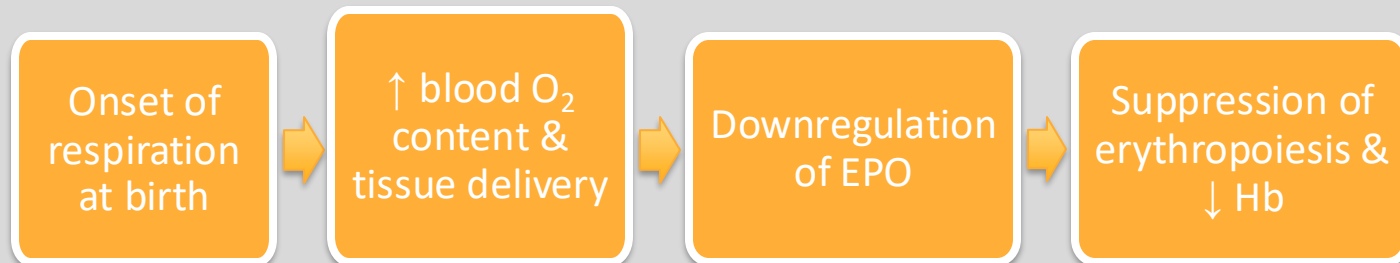
Physiologic Anemia Of Infancy/ Physiologic Nadir

Decline of Hb persisting for first 6-8 weeks of life (up to 12 wks)

Initial Decrease EPO will self correct over time and Hb returns to normal

Hb levels: 10-11 g/dL

Self-limited: no treatment or lab work needed



Transient Erythroblastopenia of Childhood

- Most common **acquired red cell aplasia** in children
- **Transient suppression** of RBC production → reticulocytopenia → normocytic anemia

Hb 6-8 g/dL but WBC and Plts are normal

Clinical presentation:

Previously healthy children between 6 mo - 3 yrs of age

Often following viral illness 2-3 months prior

Slowly developing pallor, slow recovery back to nl Hb

Treatment: none, self-limited, 1-2 month complete recovery

NO OTHER RED FLAGS!!!

Red Flags: Chronic bone pain, HSM, other cell lines affected, petechiae, signs of hemolysis

IDA Labs and Histology

- CBC with differential
 - Decreased Hb
 - Decreased MCV (microcytic)
 - High RDW
 - Thrombocytosis common
- Iron studies
 - Low Ferritin*
 - Low serum iron
 - High TIBC
 - Decreased transferrin saturation
- Normal Lead values

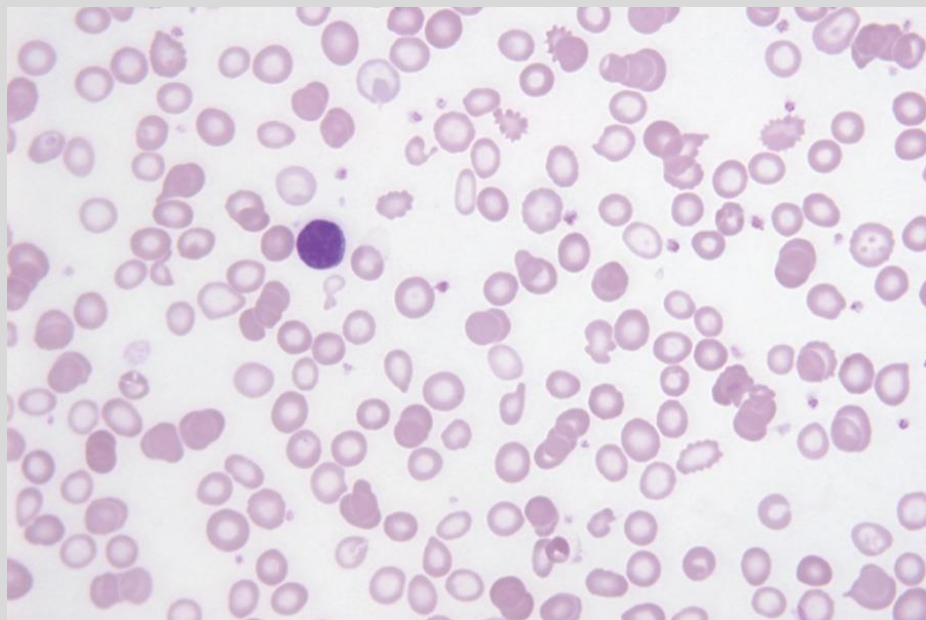


Fig. 482.1 Peripheral blood smear in iron deficiency.

Note the small, pale (microcytic, hypochromic) red blood cells with variable sizes and shapes (anisopoikilocytosis). Occasional target cells with central hemoglobin pooling, as well as several somewhat elongated hypochromic microcytes (pencil cells), are present.

(From Fleming MD: Disorders of iron and copper metabolism, the sideroblastic anemias, and lead toxicity. In Orkin SH, Nathan DG, Ginsburg D, et al, editors: Nathan and Oski's hematology and oncology of infancy and childhood, ed 8, Philadelphia, 2015, Elsevier, Fig 11-7.)

Image from: Kliegman, Robert. *Nelson Textbook of Pediatrics*. 21st ed. Elsevier, Inc, 2020. Chapter 482, p. 2523

Sickle Cell Anemia

- **Genetics:**
 - Autosomal Recessive
 - Point mutation: “GV6” Glutamine (Hydrophilic) → Valine (Hydrophobic) 6th residue in Hb β -chain results in HbS
 - Non-polar → polar
- **Pathophysiology**
 - Deoxygenated HbS interact and form rigid polymers → sickle shape
 - Sickle shape leads to microvascular vaso-occlusion in postcapillary venules
- **Diagnosis/Lab Findings:**
 - Newborn Screening
 - Hemoglobin electrophoresis
 - Blood smear: sickles, Howell jolly-bodies, target cells

Sickle cell Chest Crisis

- New pulmonary infiltrate + respiratory symptoms

Leading cause of death in SCD

Triggers

- **Infection**, fat embolism, infarction

Symptoms

- Fever, cough, chest pain, **hypoxia**
- Respiratory distress

CXR

- New infiltrate (lobar or multilobar)

Management

- **Oxygen + antibiotics**
- **Exchange transfusion** if severe



Hemorrhagic Disease of the Newborn

A bleeding disorder in newborns caused by **vitamin K deficiency**, leading to impaired coagulation. AKA Vit K deficient bleeding

Newborns have **low vitamin K stores** at birth.

Poor placental transfer, sterile gut (no vitamin K–producing flora), and immature liver

Other risk factor: infants of mothers on anticonvulsants or TB drugs.

mother on anticonvulsants (phenobarbital, phenytoin) interferes with Vit K fxn.

Prevented by receiving Vitamin K injection at birth (standard of care) –life saving prophylaxis

Classic: Week 1- bleeding from umbilicus, GI tract, circumcision.

Later onset: 2–12 weeks — may present with **intracranial hemorrhage**

Labs: **Prolonged PT**, normal platelets and PTT

Do.

Make.

Innovate.

Reinvent the Future.

A 3 year old boy is brought into the clinic due to a rash for 3 days, mom says it is spreading. No recent fever but had a viral illness recovered about 3 weeks ago. Mom notes he has been bruising more easily and has had a nose bleed today which stopped with some packing. No family Hx of any bleeding issues

On exam VSS

Skin: macular, scattered non blanching rash to the face, chest and slightly to the arms and legs. None to hands and feet.

Chest clear, CVS normal

Ab: soft, no Hepatosplenomegaly

CBC WBC 8.0, Hb 12.5 Plts 20K (nl 150-450K)

Chem panel nl

What is the most likely diagnosis?

Immune Thrombocytopenic Purpura(ITP)

- Most Common cause of Acute Onset of Thrombocytopenia
- AKA Idiopathic Thrombocytopenic Purpura

Laboratory findings:

thrombocytopenia only
(often $<20,000/\mu\text{L}$), normal Hb
and WBC count, increased
megakaryocytes on bone
marrow (usually not performed)

- **Presentation: (acute/sudden onset)**
 - **Petechiae and purpura**
 - Skin
 - Mucous Membranes
 - **Self limited most of the time**
 - **Mucosal bleeding**
 - Epistaxis (nosebleeding)
 - GI Bleed
 - Severe bleeding, such as intracranial hemorrhage, is rare

ITP Treatment

- Observation
 - **typically self-resolves**
- IVIG
 - Appears to reduce Fc-mediated phagocytosis of antibody bound platelets
- Corticosteroids
 - Short course until rise in platelet count > 20,000
- Rituximab
- Splenectomy
 - Refractory, chronic, or severe cases
- Thrombopoietin receptor agonists

Table 511.3

Treatment Options for Idiopathic Thrombocytopenic Purpura (ITP)

Adapted from Flood VH, Scott JP: Bleeding and thrombosis. In Kliegman R, Lye P, Bordini B, et al, editors: *Nelson pediatric symptom-based diagnosis*, Philadelphia, 2017, Saunders Elsevier.

	PROS	CONS	COST
Observation	Does not expose patient to unnecessary medications	May increase parent and physician anxiety	Relatively inexpensive
IVIG	Rapid response in most cases	IV administration, side effects	Expensive
Corticosteroids	Oral, effective in 70–80% of patients, minimal side effects with short courses	Side effects, may not affect long term outcome	Inexpensive
Rituximab	Long-term remission in 40–60% of patients	IV administration, immune suppression, potential for reactivation of hepatitis	Very expensive
Splenectomy	Curative in 80% of patients	Requires surgery and anesthesia, lifelong risk of infection	Expensive
Thrombopoietin receptor agonists	Potential for oral administration, 40–60% of patients respond	Not curative, usually required long term, can cause elevated liver enzymes	Very expensive

Image: Kliegman, Robert. *Nelson Textbook of Pediatrics*. 21st ed. Elsevier, Inc, 2020. Chapter 511, p. 2609-2618

Typically no Platelet transfusions!

Hemophilia Presentation

- First sign: severe/ persistent bleeding after circumcision
- **Hemarthrosis** is a hallmark
 - Complaint of a warm tingling sensation in joint
 - **Iliopsoas muscle: Large blood volume accumulation → verge on hemorrhagic shock. Can manifest as vague groin pain**
- Petechiae uncommon!



Hemophilia

- **Diagnosis/Lab Findings**
 - PTT prolonged
 - Normal PT
 - Decreased Factor Assays
 - Severe hemophilia <1% factor activity
 - Moderate hemophilia factor levels of 1–5%
 - Mild hemophilia factor levels >5%,
 - Normal Platelet count (no mucosal bleeding)

Von Willebrand Disease

- Most common inherited bleeding disorder (genetics: most AD)
- **Caused by a defect in VWF** (aids in platelet adhesions and protecting factor VIII)
- Many different types; may be a quantitative or qualitative defect
 - Most common is type 1: mild to moderate quantitative defect
- **Clinical presentation**
 - Mucocutaneous bleeding (epistaxis, bleeding gums)
 - Postoperative bleedings (eg after a dental extraction)
 - **Adolescents: excessive menstrual bleeding**
- **Diagnosis:** VWF antigen, ristocetin cofactor activity, factor VIII level, screening test not always reliable (may need to rpt)
 - **Lab values:** Normal PT, ~ increased PTT (depending on FVIII levels), increased bleeding time
- **Treatment:** Desmopressin (DDAVP), Fresh frozen plasma

Suspecting Leukemia

- **Early—nonspecific**
 - Anorexia (unintentional weight loss), fatigue, malaise, irritability, and intermittent low-grade **fever**.
 - **Chronic Bone/joint pain**, limp- (don't want to play/ normal activities red flag)
- **Late—bone marrow failure**
 - **Pallor**, fatigue, exercise intolerance--anemia
 - **Petechiae**, ecchymoses, oral mucosal bleeding or epistaxis--thrombocytopenia
 - Persistent **fever** (infections)--neutropenia
- **Organ Infiltration**
 - Lymphadenopathy, **Hepatosplenomegaly**, Testicular enlargement
 - Central nervous system (CNS) involvement (headache, seizures)
 - “Blueberry muffin” appearance in AML, DIC*

Acute Leukemia Treatment

- **Chemotherapy**
 - Risk stratification
 - Age, WBC at diagnosis, immunophenotype and cytogenetics, response to initial treatment
 - Intrathecal for CNS
 - **Presence of genetic disorders (Down syndrome) can influence treatment**
- **Relapses**
 - Cranial or craniospinal irradiation in later phases if high risk of CNS relapse
 - Hematopoietic Stem Cell Transplant (HSCT) for patients with relapse- improves long term survival
- **Watch For:**
 - Tumor Lysis Syndrome
 - Myelosuppression and risk for infection

Emergency Complications to watch for:

- Hyperleukocytosis
 - WBC > 100,000/mL³
 - → white thrombi of leukocyte aggregates
→ microvascular injury to lungs and CNS
 - Hyperviscosity, sludging
- Metabolic derangements d/t
 - Disseminated intravascular coagulation (DIC)
 - **Electrolyte abnormalities from tumor lysis syndrome (renal, cardiac, neuro)**
 - Rapid cell turnover → hyperuricemia, hyperkalemia, hyperphosphatemia, hypocalcemia
- Signif Anemia
- Thrombocytopenia (Bleeding risk)
- **Infection or neutropenia (sepsis)**
 - **Fever, sepsis, bacteremia**
- Airway compression or superior vena cava syndrome (SVC)
 - Anterior mediastinal mass
 - Respiratory distress, cough, orthopnea
 - Headache, syncope
 - Facial swelling, plethora

Neuroblastoma

- 90% diagnosed <5 yo, median age 22 mo
- **Derived from primordial neural crest cells (can arise in any peripheral sympathetic tissue)**
- Presentation depends on site of the tumor and extent:
 - Abdomen/adrenal- **abdominal mass** or bowel obstruction
 - Thorax- stridor, resp distress, vena cava syndrome
 - Cervical: Horner syndrome
 - Paraspinal- spinal cord compression

Neuroblastoma Presentation

Paraneoplastic Syndromes:

- **Opsoclonus-Myoclonus-Ataxia-Syndrome**
- **(Dancing eyes/ Dancing Feet)**
 - Rapid, uncontrollable jerking eye and body movements, poor coordination, and cognitive dysfunction
- Vasoactive Intestinal Peptide Secretion
 - Profound secretory diarrhea
- Catecholamine secretion
 - Sweating and HTN

Diagnosis

- Abdominal mass on Xray/MRI/CT
- Homer-Wright Rosettes, small round blue cells on bone marrow aspirate
- **Catecholamine metabolites**
 - **Urine HVA and VMA**
(homovanillic acid and vanillylmandelic acid)

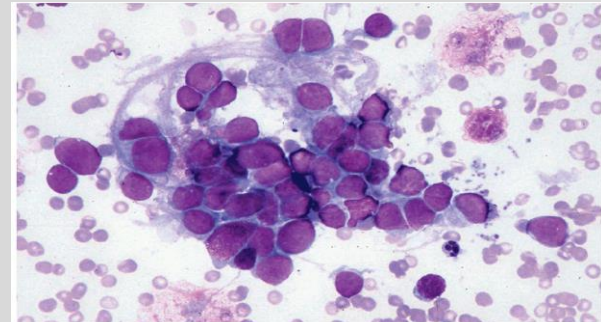


Fig. 525.4

Neuroblastoma cells aspirated from the bone marrow. Clumps of cells often contain 3 or more cells with or without evidence of rosette formation. Rosettes of cells surrounding an inner mass of fibrillary material are characteristic of neuroblastoma.

Wilms Tumor Presentation

- **Asymptomatic abdominal mass**
 - Most of the time U/L and does not cross midline
- HTN sometimes (25%)
- Abdominal pain possible
- Gross painless hematuria

Most Common renal malignancy in children

75% of cases < 5 yo, peak 2-3 yo

- Most cases are sporadic, 2% are familial.

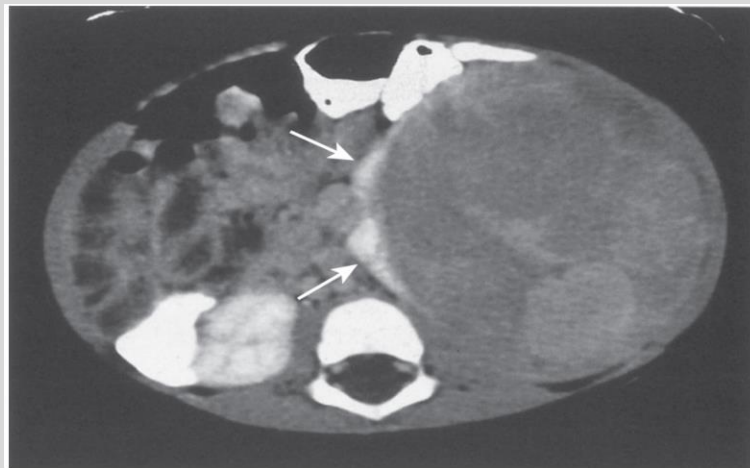


Fig. 526.1

Large heterogeneous Wilms tumor in left kidney.

The residual renal parenchyma is displaced medially (arrows).

(From Haaga JR, Boll DT, editors: *CT and MRI of the whole body*; vol II, ed 6, Philadelphia, 2017, Elsevier, Fig 54-84, p 1816.)

Wilms Tumor Diagnosis and Treatment

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HALLMARK: ASYMPTOMATIC ABDOMINAL MASS incidentally found on exam by parent or physician

Diagnosis:

- Abdominal exam: firm, nontender, not crossing midline
- Xray/CT/MRI/US
- Histology: blastemal, stromal, and epithelial cells
- Metastases to lungs, lymph nodes, liver

Treatment:

- Nephrectomy
- Chemotherapy
- Radiation
- Good prognosis >90%



Retinoblastoma

- Most common intraocular tumor in children
- ~250-350 new cases of retinoblastoma each year in US—median age ~2 yr
- **Genetics:**
 - RB1—tumor suppressor gene, 2 hit hypothesis
- **Presentation**
 - Absent or abnormal red reflex
 - Leukocoria
 - Strabismus
 - Decreased vision
 - Pupil irregularity

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Image: Kliegman, Robert. *Nelson Textbook of Pediatrics*. 21st ed. Elsevier, Inc, 2020.
Chapter 648, p. 3377-3385

Retinoblastoma

- **Diagnosis:**
 - Retinal mass on ophthalmologic exam performed under anesthesia
 - Histology: small round blue cell tumor with rosette formation
- **Treatment:**
 - Enucleation if vision cannot be spared
 - B/L: chemoreduction and focal therapy (laser photocoagulation or cryotherapy)
 - Small tumor: focal therapy
 - Careful follow-up

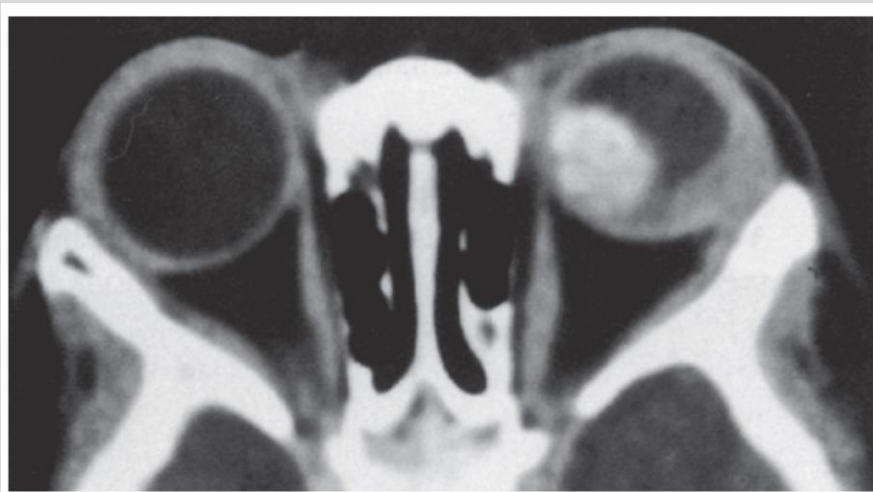


Fig. 529.2

Axial contrast-enhanced CT scan shows calcified retinoblastoma of the left eye.

(From Haaga JR, Boll DT, et al, editors: *CT and MRI of the whole body*, ed 6, Philadelphia, 2017, Elsevier, Fig 20-32.)

Image: Kliegman, Robert. *Nelson Textbook of Pediatrics*. 21st ed. Elsevier, Inc, 2020. Chapter 529, p. 2697-2698

Ewing Sarcoma

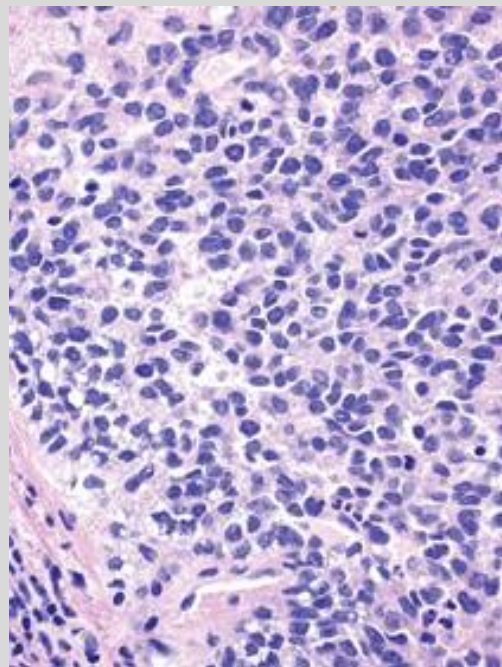
- Malignant neoplasm of primitive neuroectodermal cells
- 2.1 cases per 1 million children in US
- Caucasian predominance
- **Genetics:**
 - t(11;22)—EWSR1 (22) FLI1 gene (11)
- **Presentation**
 - Pain, swelling, limited ROM
 - Bone pain worse at night
 - Tenderness over involved bone
 - Early Metastasis common (lung)
 - **Fever, leukocytosis, weight loss (systemic signs)**
 - Can be mistaken for sports injury.
 - Minor trauma, pain intensifies rapidly

Ewing Sarcoma

- **Diagnosis/Lab Findings:**
 - **“Onion Skinning”**- periosteal rxn building and destruction bone layers
 - Histology: sheets of primitive small, round blue cells
 - Long bones or flat bones
 - Diaphysis
- **Treatment:**
 - Rule out osteomyelitis or secondary bone tumor
 - Surgery, chemotherapy, and radiation



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Osteosarcoma

- Malignant bone neoplasm of osteoblasts
- 5.6 cases per 1 million children <15 yr old in US
- **Genetics:**
 - Rb—retinoblastoma
 - P53—Li-Fraumeni
- **Presentation:**
 - Highest risk period during adolescent growth spurt
 - **Pain, limp, and swelling are most common presentation- esp worsening/ persistent**
 - Pathologic fracture
 - Limited ROM, joint effusion, tenderness, warmth
 - Usually no other systemic signs

Osteosarcoma

- **Diagnosis/Lab Findings**
 - Long Bones:
 - Metaphysis
 - Knee
 - Biopsy
 - Definitive diagnosis, pleomorphic osteoid-producing cells
 - Imaging:
 - **Sunburst Pattern**, Codman's triangle (periosteal elevation)
- **Treatment**
 - Surgical resection + chemotherapy
 - 65-75% 5 year survival for nonmetastatic disease



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